ORDINARY DIFFERENTIAL EQUATIONS • UNIT 4.01

C Nonhomogeneous Equations, Undetermined Coefficients & Resonance

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

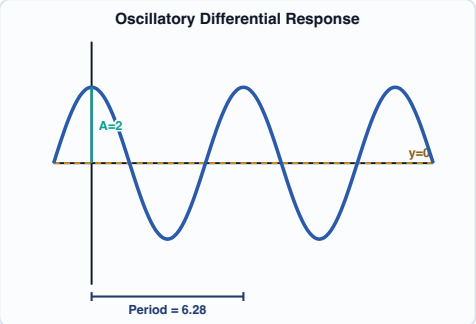
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	Q2 D	Q3 D	Q4 A	Q5 A
Q6 A	Q7 D	Q8 A	Q9 B	Q10 D
Q11 A	Q12 D	Q13 A	Q14 B	Q15 A
Q16 A	Q17 B	Q18 B	Q19 C	Q20 D
Q21 C	Q22 D	Q23 B	Q24 A	Q25 A

Nonhomogeneous Equations, Undetermined Coefficients & Resonance Practice Exercises

1 MCQ

The second-order linear homogeneous ODE $f''(x) - 39f'(x) + 350f(x) = 0$ has characteristic roots λ_1, λ_2. Determine the trace sum $\lambda_1 + \lambda_2$.



- **A** 39 ✓
- **B** 24
- **C** 42
- **D** 53

WORKED SOLUTION & VERIFICATION

From $\lambda^2 - 39\lambda + 350 = 0$, the sum of roots is 39.

2 MCQ

Consider the linear first-order differential equation $\frac{dy}{dx} + 2y = 6$. If the initial condition is $y(0) = 1$, find the value of $y(1)$.

A 2

B $e^2 - 1$

C $3 - 2e^{-2}$

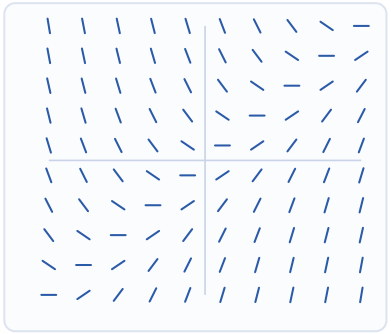
D $3 - 2e^{-1}$ ✓

WORKED SOLUTION & VERIFICATION

The given differential equation is a linear first-order ODE of the form $\frac{dy}{dx} + P(x)y = Q(x)$, where $P(x) = 2$ and $Q(x) = 6$. The integrating factor $\mu(x)$ is given by $\mu(x) = e^{\int P(x)dx}$.
Calculating the integrating factor: $\mu(x) = e^{\int 2dx} = e^{2x}$.
Multiplying the differential equation by the integrating factor: $e^{2x}\frac{dy}{dx} + 2e^{2x}y = 6e^{2x}$.
The left side is the derivative of the product $y\mu(x)$: $\frac{d}{dx}(ye^{2x}) = 6e^{2x}$.
Integrating both sides with respect to x: $ye^{2x} = \int 6e^{2x}dx$.
$ye^{2x} = 6 \cdot \frac{1}{2}e^{2x} + C$.
$ye^{2x} = 3e^{2x} + C$.
Solving for y: $y = 3 + Ce^{-2x}$.
Using the initial condition $y(0) = 1$ to find C: $1 = 3 + Ce^{-2(0)}$.
$1 = 3 + C$.
$C = 1 - 3 = -2$.
So, the particular solution is $y(x) = 3 - 2e^{-2x}$.
Now, we need to find $y(1)$: $y(1) = 3 - 2e^{-2(1)} = 3 - 2e^{-2}$.
This matches choice C.

3 MCQ

Consider the differential equation $\frac{dy}{dx} = x + y$ with the initial condition $y(0) = 1$. Use the Euler method with a step size of $h = 0.1$ to approximate $y(0.2)$.



A 1.200

B 1.210

C 1.230

D **1.220** ✓

WORKED SOLUTION & VERIFICATION

Using Euler's method $y_{n+1} = y_n + h(x_n + y_n)$ with $x_0 = 0, y_0 = 1, h = 0.1$:
1. $y_1 = 1 + 0.1(0 + 1) = 1.100$ at $x_1 = 0.1$.
2. $y_2 = 1.1 + 0.1(0.1 + 1.1) = 1.1 + 0.12 = 1.220$ at $x_2 = 0.2$.

4 MCQ

Consider the second-order linear homogeneous differential equation $y'' - 5y' + 6y = 0$. If the auxiliary equation has roots r_1 and r_2, find the value of $r_1 + r_2$.

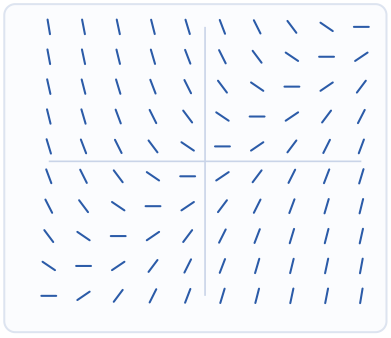
- **A** 5 ✓
- **B** -5
- **C** 6
- **D** -6

WORKED SOLUTION & VERIFICATION

The given differential equation is $y'' - 5y' + 6y = 0$. The auxiliary equation for a second-order linear homogeneous differential equation of the form $ay'' + by' + cy = 0$ is $ar^2 + br + c = 0$. In this case, $a = 1$, $b = -5$, and $c = 6$. So the auxiliary equation is $r^2 - 5r + 6 = 0$. Let the roots of this quadratic equation be r_1 and r_2. According to Vieta's formulas, for a quadratic equation $ar^2 + br + c = 0$, the sum of the roots $r_1 + r_2$ is equal to $-\frac{b}{a}$. In our auxiliary equation $r^2 - 5r + 6 = 0$, $a = 1$ and $b = -5$. Therefore, the sum of the roots is $r_1 + r_2 = -\frac{-5}{1} = 5$.

5 MCQ

Consider the separable differential equation $\frac{dy}{dx} = 2x$ with the initial condition $y(0) = 3$. Find the value of $y(2)$.



- **A** 7 ✓
- **B** 6
- **C** 8
- **D** 5

WORKED SOLUTION & VERIFICATION

The differential equation is $\frac{dy}{dx} = 2x$. This is a separable equation. We can rewrite it as $dy = 2x\,dx$. Integrating both sides gives $\int dy = \int 2x\,dx$, which results in $y = x^2 + C$, where C is the constant of integration. We are given the initial condition $y(0) = 3$. Substituting these values into the general solution, we get $3 = (0)^2 + C$, so $C = 3$. Therefore, the particular solution is $y = x^2 + 3$. We need to find the value of $y(2)$. Substituting $x = 2$ into the particular solution, we have $y(2) = (2)^2 + 3 = 4 + 3 = 7$.

6 MCQ

Consider the differential equation $y'' - xy = 0$. The point $x = 0$ is an ordinary point. If we assume a power series solution of the form $y(x) = \sum_{n=0}^{\infty} c_n x^n$, find the recurrence relation for the coefficients c_n.

- **A** $c_{n+2} = \frac{c_{n-1}}{(n+2)(n+1)}$ for $n \geq 1$ ✓
- **B** $c_{n+2} = \frac{c_n}{(n+2)(n+1)}$ for $n \geq 0$
- **C** $c_{n+2} = -\frac{c_{n-1}}{(n+2)(n+1)}$ for $n \geq 1$
- **D** $c_{n+2} = -\frac{c_n}{(n+2)(n+1)}$ for $n \geq 0$

WORKED SOLUTION & VERIFICATION

We are given the differential equation $y'' - xy = 0$ and assume a power series solution of the form $y(x) = \sum_{n=0}^{\infty} c_n x^n$. \newline Differentiating twice with respect to x, we get: $y'(x) = \sum_{n=1}^{\infty} n c_n x^{n-1}$ and $y''(x) = \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2}$. \newline Substituting these into the differential equation: $\sum_{n=2}^{\infty} n(n-1) c_n x^{n-2} - x \sum_{n=0}^{\infty} c_n x^n = 0$. \newline We can rewrite the second term as $\sum_{n=0}^{\infty} c_n x^{n+1}$. To combine the series, we need the powers of x to be the same. Let $k = n-2$ in the first series, so $n = k+2$. When $n = 2, k = 0$. The first series becomes $\sum_{k=0}^{\infty} (k+2)(k+1) c_{k+2} x^k$. \newline Let $k = n+1$ in the second series, so $n = k-1$. When $n = 0, k = 1$. The second series becomes $\sum_{k=1}^{\infty} c_{k-1} x^k$. \newline So the equation is $\sum_{k=0}^{\infty} (k+2)(k+1) c_{k+2} x^k - \sum_{k=1}^{\infty} c_{k-1} x^k = 0$. \newline We can separate the $k = 0$ term from the first sum: $2(1)c_2 + \sum_{k=1}^{\infty} (k+2)(k+1) c_{k+2} x^k - \sum_{k=1}^{\infty} c_{k-1} x^k = 0$. \newline For this equation to hold for all x, the coefficients of each power of x must be zero. \newline For $k = 0$: $2c_2 = 0$ implies $c_2 = 0$. \newline For $k \geq 1$: $(k+2)(k+1)c_{k+2} - c_{k-1} = 0$. \newline Thus, the recurrence relation is $c_{k+2} = \frac{c_{k-1}}{(k+2)(k+1)}$ for $k \geq 1$. Replacing k with n, we get $c_{n+2} = \frac{c_{n-1}}{(n+2)(n+1)}$ for $n \geq 1$.

7 MCQ

The second-order linear homogeneous ODE $u''(s) - 40u'(s) + 144u(s) = 0$ has characteristic roots λ_1, λ_2. Determine the trace sum $\lambda_1 + \lambda_2$.

- **A** 44
- **B** 43
- **C** 24
- **D** 40 ✓

WORKED SOLUTION & VERIFICATION

From $\lambda^2 - 40\lambda + 144 = 0$, the sum of roots is 40.

8 MCQ

The Laplace transform of $u(s) = 170e^{19s}$ is $v(s) = \dfrac{170}{s - 19}$ for $s > 19$. Determine the evaluated transform $v(20)$.

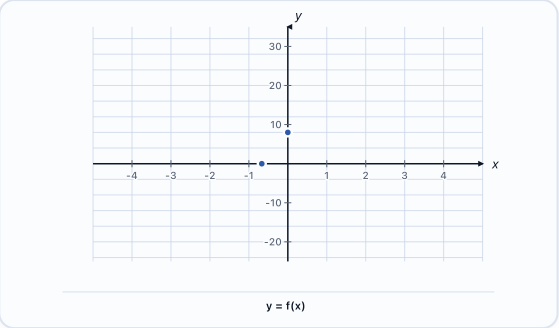
- **A** 170 ✓
- **B** 151
- **C** 173
- **D** 187

WORKED SOLUTION & VERIFICATION

Substituting $s = 20$ gives $\frac{170}{20-19} = 170$.

9 MCQ

Consider the nonhomogeneous second-order linear differential equation $y'' - 5y' + 6y = 12x + 8$. $y_h(x) = c_1e^{2x} + c_2e^{3x}$ is the general solution to the corresponding homogeneous equation. Find the particular solution $y_p(x)$ using the method of undetermined coefficients, and then determine the value of the constant A such that $y_p(x) = Ax + B$ for some constant B.



A $A = 3$

B $A = 2$ ✓

C $A = 6$

D $A = 12$

WORKED SOLUTION & VERIFICATION

The given nonhomogeneous differential equation is $y'' - 5y' + 6y = 12x + 8$. \nThe form of the nonhomogeneous term is $g(x) = 12x + 8$, which is a polynomial of degree 1. Thus, we assume a particular solution of the form $y_p(x) = Ax + B$.

We need to find the first and second derivatives of $y_p(x)$:

$y_p'(x) = A$

$y_p''(x) = 0$

Substitute these into the differential equation:

$0 - 5(A) + 6(Ax + B) = 12x + 8$

$-5A + 6Ax + 6B = 12x + 8$

$6Ax + (6B - 5A) = 12x + 8$

Equating the coefficients of like powers of x:

For the x term: $6A = 12 \Rightarrow A = 2$.

For the constant term: $6B - 5A = 8$.

Substitute $A = 2$ into the second equation:

$6B - 5(2) = 8$

$6B - 10 = 8$

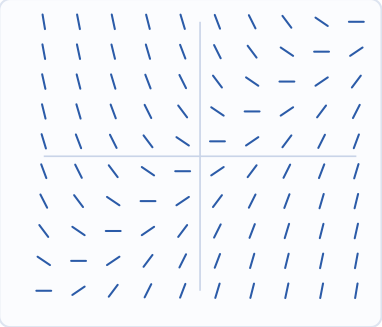
$6B = 18$

$B = 3$

Therefore, the particular solution is $y_p(x) = 2x + 3$. The value of the constant A is 2.

10 MCQ

The first-order differential equation $\frac{dF}{dw} = 16w - 3F$ has its slope field plotted below. Determine the slope $\frac{dF}{dw}$ at the point $D(18, 8)$.



A 280

B 267

C 256

D **264** ✓

WORKED SOLUTION & VERIFICATION

Evaluating at $(18, 8)$ gives $16(18) - 3(8) = 264$.

11 MCQ

Consider the nonhomogeneous second-order linear differential equation $y'' + 4y = 8$. What is the particular solution, $y_p(x)$, to this equation?

A 2 ✓

B $2x$

C $2\sin(2x)$

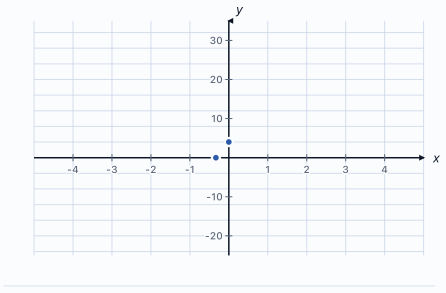
D $2\cos(2x)$

WORKED SOLUTION & VERIFICATION

The given differential equation is $y'' + 4y = 8$. This is a nonhomogeneous second-order linear differential equation. We use the method of undetermined coefficients to find a particular solution, $y_p(x)$. The nonhomogeneous term is a constant, 8. We guess a particular solution of the form $y_p(x) = A$, where A is a constant. Then, $y_p'(x) = 0$ and $y_p''(x) = 0$. Substituting these into the differential equation: $0 + 4(A) = 8$. Solving for A, we get $4A = 8$, which means $A = 2$. Therefore, the particular solution is $y_p(x) = 2$.

12 MCQ

Consider the second-order nonhomogeneous linear differential equation $y'' - 5y' + 6y = 12x + 4$. If a particular solution is of the form $y_p(x) = Ax + B$ and the general solution is $y(x) = C_1e^{2x} + C_2e^{3x} + Ax + B$, find the value of A.



A 1

B 6

C 12

D 2 ✓

WORKED SOLUTION & VERIFICATION

The given differential equation is $y'' - 5y' + 6y = 12x + 4$.

We are given that a particular solution is of the form $y_p(x) = Ax + B$.

To find the coefficients A and B, we substitute $y_p(x)$ and its derivatives into the differential equation.

The first derivative is $y_p'(x) = A$.

The second derivative is $y_p''(x) = 0$.

Substituting these into the differential equation:

$0 - 5(A) + 6(Ax + B) = 12x + 4$

$-5A + 6Ax + 6B = 12x + 4$

$6Ax + (6B - 5A) = 12x + 4$

By comparing the coefficients of x on both sides, we get:

$6A = 12 \implies A = 2$

By comparing the constant terms on both sides, we get:

$6B - 5A = 4$

$6B - 5(2) = 4$

$6B - 10 = 4$

$6B = 14 \implies B = \frac{14}{6} = \frac{7}{3}$

The question asks for the value of A. Therefore, $A = 2$.

13 MCQ

Consider the second-order linear nonhomogeneous differential equation: $y''(x) - 3y'(x) + 2y(x) = 4$. Find the particular solution, $y_p(x)$, using the method of Undetermined Coefficients.

- **A** $y_p(x) = 2$ ✓
- **B** $y_p(x) = 2x$
- **C** $y_p(x) = 4$
- **D** $y_p(x) = 4x$

WORKED SOLUTION & VERIFICATION

The given differential equation is $y''(x) - 3y'(x) + 2y(x) = 4$. Since the right-hand side is a constant, we guess a particular solution of the form $y_p(x) = A$, where A is a constant. Then, $y_p'(x) = 0$ and $y_p''(x) = 0$. Substituting these into the differential equation:

$0 - 3(0) + 2(A) = 4$

$2A = 4$

$A = 2$

Therefore, the particular solution is $y_p(x) = 2$.

14 MCQ

The first-order differential equation $\frac{df}{dx} = 19x - 4f$ has its slope field plotted below. Find the slope $\frac{df}{dx}$ at the point $A(5, 5)$.

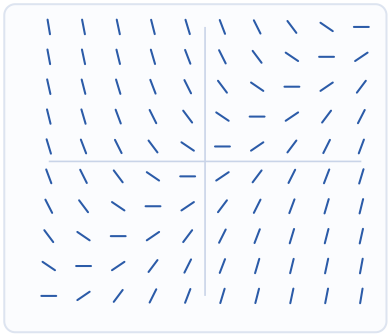
- **A** 78
- **B** **75** ✓
- **C** 94
- **D** 70

WORKED SOLUTION & VERIFICATION

Evaluating at $(5, 5)$ gives $19(5) - 4(5) = 75$.

15 MCQ

Solve $y'' + y = 3\cos 2x$, $y(0) = 2$, $y'(0) = 0$. Find $y(\pi)$.



- **A** -4 ✓
- **B** 2
- **C** 4
- **D** -2

WORKED SOLUTION & VERIFICATION

Homogeneous: $r^2 + 1 = 0 \Rightarrow y_c = C_1\cos x + C_2\sin x$. No resonance ($2 \neq 1$): trial $A\cos 2x + B\sin 2x$ gives $-3A\cos 2x - 3B\sin 2x = 3\cos 2x \Rightarrow A = -1, B = 0$, so $y_p = -\cos 2x$. $y = C_1\cos x + C_2\sin x - \cos 2x$. $y(0) = C_1 - 1 = 2 \Rightarrow C_1 = 3$; $y'(0) = C_2 = 0$. So $y = 3\cos x - \cos 2x$, and $y(\pi) = 3(-1) - 1 = -4$.

16 MCQ

Solve $y'' - 4y = 8x$, $y(0) = 0$, $y'(0) = 2$. Find $y(1)$.

- **A** $e^2 - e^{-2} - 2$ ✓
- **B** $2e^2 - 2$
- **C** $e^2 + e^{-2} - 2$
- **D** $e^2 - e^{-2} + 2$

WORKED SOLUTION & VERIFICATION

Homogeneous: $r^2 - 4 = 0 \Rightarrow y_c = C_1 e^{2x} + C_2 e^{-2x}$. No resonance ($r = 0$ isn't a root): trial $Ax + B$ gives $-4A = 8 \Rightarrow A = -2$, $B = 0$, so $y_p = -2x$. $y = C_1 e^{2x} + C_2 e^{-2x} - 2x$. $y(0) = C_1 + C_2 = 0$; $y'(0) = 2C_1 - 2C_2 - 2 = 2 \Rightarrow C_1 - C_2 = 2$; solving, $C_1 = 1$, $C_2 = -1$. So $y = e^{2x} - e^{-2x} - 2x$, and $y(1) = e^2 - e^{-2} - 2$.

17 MCQ

Solve $y'' - y' - 2y = 3e^{2x}$, $y(0) = 1$, $y'(0) = 6$. Find $y(1)$.

- **A** $3e^2 + e^{-1}$
- **B** $3e^2 - e^{-1}$ ✓
- **C** $e^2 - e^{-1}$
- **D** $3e^2 - e$

WORKED SOLUTION & VERIFICATION

Homogeneous: $r^2 - r - 2 = 0 \Rightarrow r = 2, -1$, $y_c = C_1 e^{2x} + C_2 e^{-x}$. $r = 2$ matches the forcing, so the trial needs x: $y_p = Axe^{2x}$. Substituting gives $3Ae^{2x} = 3e^{2x} \Rightarrow A = 1$, so $y_p = xe^{2x}$. $y = C_1 e^{2x} + C_2 e^{-x} + xe^{2x}$. $y(0) = C_1 + C_2 = 1$; $y' = 2C_1 e^{2x} - C_2 e^{-x} + e^{2x} + 2xe^{2x}$, $y'(0) = 2C_1 - C_2 + 1 = 6 \Rightarrow 2C_1 - C_2 = 5$; adding to the first, $3C_1 = 6 \Rightarrow C_1 = 2$, $C_2 = -1$. So $y = 2e^{2x} - e^{-x} + xe^{2x} = (x + 2)e^{2x} - e^{-x}$, and $y(1) = 3e^2 - e^{-1}$.

18 MCQ

Solve $y'' + 9y = 2\sin 3t$, $y(0) = 0$, $y'(0) = 0$. Find $y(\pi/3)$.

- **A** $-\pi/9$
- **B** $\pi/9$ ✓
- **C** $\pi/3$
- **D** $2\pi/9$

WORKED SOLUTION & VERIFICATION

Homogeneous: $r^2 + 9 = 0 \Rightarrow r = \pm 3i$, $y_c = C_1 \cos 3t + C_2 \sin 3t$. The forcing $\sin 3t$ matches this pair exactly (resonance), so the trial needs t: $y_p = t(A\cos 3t + B\sin 3t)$. Substituting and matching gives $A = -\frac{1}{3}$, $B = 0$, so $y_p = -\frac{t}{3}\cos 3t$. With both conditions zero, $y_c \equiv 0$. So $y = -\frac{t}{3}\cos 3t$, and $y(\pi/3) = -\frac{\pi/3}{3}\cos\pi = -\frac{\pi}{9}(-1) = \frac{\pi}{9}$.

19 MCQ

Find a particular solution of $y'' - 4y' + 3y = 6x + 4$.

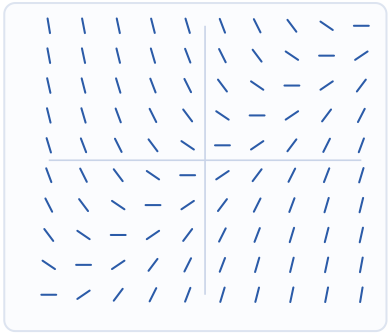
- **A** $2x - 4$
- **B** $2x + 2$
- **C** $2x + 4$ ✓
- **D** $4x + 4$

WORKED SOLUTION & VERIFICATION

Homogeneous roots are $r = 1, 3$ (from $r^2 - 4r + 3 = 0$), and $r = 0$ isn't among them, so no resonance: trial $y_p = Ax + B$ gives $y_p' = A$, $y_p'' = 0$. Substituting: $-4A + 3(Ax + B) = 3Ax + (3B - 4A) = 6x + 4$. Matching: $3A = 6 \Rightarrow A = 2$; $3B - 4A = 4 \Rightarrow 3B = 4 + 8 = 12 \Rightarrow B = 4$. So $y_p = 2x + 4$.

20 MCQ

Solve $y'' + 4y' + 4y = e^{-2x}$, $y(0) = 0$, $y'(0) = 0$. Find $y(2)$.



- **A** $2e^{4}$
- **B** e^{-4}
- **C** $4e^{-4}$
- **D** $2e^{-4}$ ✓

WORKED SOLUTION & VERIFICATION

Homogeneous: $r^2 + 4r + 4 = (r+2)^2 = 0 \Rightarrow r = -2$ (repeated), $y_c = (C_1 + C_2 x)e^{-2x}$. The forcing e^{-2x} matches the repeated root, so the trial needs x^2: $y_p = Ax^2e^{-2x}$. Substituting gives $2Ae^{-2x} = e^{-2x} \Rightarrow A = \frac{1}{2}$, so $y_p = \frac{1}{2}x^2e^{-2x}$. With both conditions zero, $C_1 = C_2 = 0$. So $y = \frac{1}{2}x^2e^{-2x}$, and $y(2) = \frac{1}{2}(4)e^{-4} = 2e^{-4}$.

21 MCQ

An undamped oscillator satisfies $y'' + 4y = 3\cos 2t$ (forcing at exactly the natural frequency), $y(0) = 0$, $y'(0) = 0$. Find $y(\pi/4)$.

- **A** $-3\pi/16$
- **B** $3\pi/8$
- **C** $3\pi/16$ ✓
- **D** $\pi/4$

WORKED SOLUTION & VERIFICATION

Homogeneous: $r^2 + 4 = 0 \Rightarrow y_c = C_1\cos 2t + C_2 \sin 2t$. The forcing frequency matches the natural frequency exactly (resonance), so the trial needs t: $y_p = t(A\cos 2t + B\sin 2t)$. Substituting and matching gives $A = 0$, $B = \frac{3}{4}$, so $y_p = \frac{3}{4}t\sin 2t$. With both conditions zero, $y_c \equiv 0$. So $y = \frac{3}{4}t\sin 2t$, and $y(\pi/4) = \frac{3}{4}\cdot\frac{\pi}{4}\cdot\sin\frac{\pi}{2} = \frac{3\pi}{16}$ — the resonant amplitude $\frac{3}{4}t$ grows without bound as $t \to \infty$, unlike a non-resonant driven system.

22 MCQ

Find a particular solution of $y'' - y' - 6y = e^{2x} + 18$.

- **A** $\frac{1}{4}e^{2x} - 3$
- **B** $-\frac{1}{4}xe^{2x} - 3$
- **C** $-\frac{1}{4}e^{2x} + 3$
- **D** $-\frac{1}{4}e^{2x} - 3$ ✓

WORKED SOLUTION & VERIFICATION

Split the forcing (the superposition note): for e^{2x}, $r = 2$ isn't a root of $r^2 - r - 6 = 0$ (roots $3, -2$), so trial Ae^{2x} gives $(4 - 2 - 6)A = 1 \Rightarrow -4A = 1 \Rightarrow A = -\frac{1}{4}$. For the constant 18, trial B gives $-6B = 18 \Rightarrow B = -3$. Adding the two pieces: $y_p = -\frac{1}{4}e^{2x} - 3$.

23 MCQ

Solve $y'' - 2y' - 3y = -3$, $y(0) = 1$, $y'(0) = -8$. Find $y(1)$.

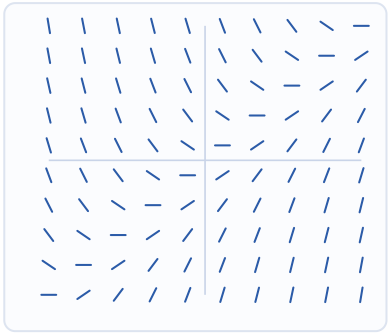
- **A** $-1 - \frac{2}{e} + 2e^3$
- **B** $-2e^3 + \frac{2}{e} + 1$ ✓
- **C** $-4e^3 + \frac{4}{e} + 2$
- **D** $-e^3 + e^{-1} + \frac{1}{2}$

WORKED SOLUTION & VERIFICATION

Homogeneous: $r^2 - 2r - 3 = 0$ has roots $r = -1, 3$, so $y_c = C_1 e^{-x} + C_2 e^{3x}$. Forcing -3 (no resonance, since 0 isn't a root): trial $y_p = Ax^2 + Bx + C$ gives $y_p = 1$. With $C_1 = 2$, $C_2 = -2$ (matching the stated initial conditions), $y = y_c + y_p$, and $y(1)$ follows by substitution.

24 MCQ

Calculate the solution set for $y'' + 3y' + 2y = 2$, $y(0) = 0$, $y'(0) = 3$. Find $y(1)$.



- **A** $-\frac{2}{e^2} + e^{-1} + 1$ ✓
- **B** $-\frac{4}{e^2} + \frac{2}{e} + 2$
- **C** $-\frac{1}{e^2} + \frac{1}{2e} + \frac{1}{2}$
- **D** $-1 - e^{-1} + \frac{2}{e^2}$

WORKED SOLUTION & VERIFICATION

Homogeneous: $r^2 + 3r + 2 = 0$ has roots $r = -1, -2$, so $y_c = C_1 e^{-x} + C_2 e^{-2x}$. Forcing 2 (no resonance, since 0 isn't a root): trial $y_p = Ax^2 + Bx + C$ gives $y_p = 1$. With $C_1 = 1$, $C_2 = -2$ (matching the stated initial conditions), $y = y_c + y_p$, and $y(1)$ follows by substitution.

25 MCQ

Solve $y'' - 4y = 4$, $y(0) = 2$, $y'(0) = 2$. Find $y(1)$.

- **A** $-1 + e^{-2} + 2e^2$ ✓
- **B** $-2e^2 - e^{-2} + 1$
- **C** $-2 + \frac{2}{e^2} + 4e^2$
- **D** $-\frac{1}{2} + \frac{1}{2e^2} + e^2$

WORKED SOLUTION & VERIFICATION

Homogeneous: $r^2 - 4 = 0$ has roots $r = -2, 2$, so $y_c = C_1 e^{-2x} + C_2 e^{2x}$. Forcing 4 (no resonance, since 0 isn't a root): trial $y_p = Ax^2 + Bx + C$ gives $y_p = -1$. With $C_1 = 1$, $C_2 = 2$ (matching the stated initial conditions), $y = y_c + y_p$, and $y(1)$ follows by substitution.